



Solving the Lorenz ODE System using **Optimal ANN** Architectures

Can a **plain feedforward ANN** — no physics in the loss — learn a coupled nonlinear ODE system, and **which architecture does it best?**

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Final Year Design Project I

Introduction & Motivation

⚡ Classical Solvers

Highly accurate but strictly **mesh-dependent**. Re-integrates step-by-step for every query, causing large computational overhead.

🌈 Neural-ODE Methods

Maps inputs continuously in one shot, but typically depends on **physics-informed losses** or complex operator learning.

🔍 Plain-ANN Approach

Can a **plain feedforward ANN** (no physics in the loss) approximate the Lorenz-1960 solution purely from numerical data?

Central Question: Can a plain feedforward ANN approximate the Lorenz-1960 solution, and **which architecture performs best?**

Objectives

Dual-Solver Reference

Build a trusted reference with **RK4** and **SciPy DOP853** solvers, cross-validating accuracy.

01

Systematic Search

Systematically vary **depth**, **width**, and **activation** functions for plain ANNs (no physics aids).

02

Performance Scoring

Score performance vs cost: evaluate MSE, MAE, max error, and training/inference time.

03

PINN Comparison

Benchmark the best plain ANN configuration against published **PINN** / **DeepONet** results.

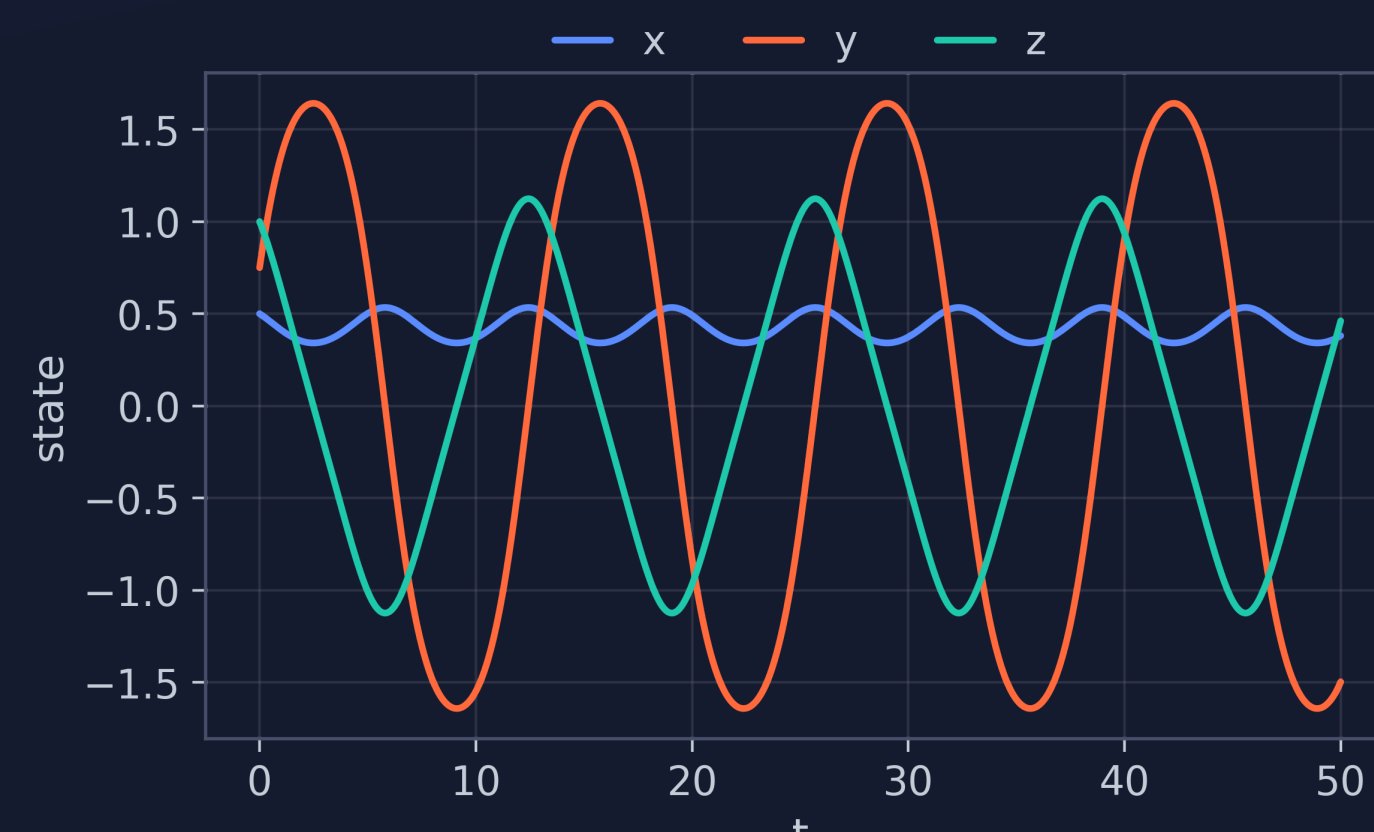
04

Background: The System

$$\begin{aligned}\dot{x} &= -0.10 \, y \, z \\ \dot{y} &= +1.60 \, x \, z \\ \dot{z} &= -0.75 \, x \, y\end{aligned}$$

$k = 2, \ell = 1 \cdot \quad x_0=0.5, \, y_0=0.75, \, z_0=1 \quad \cdot \quad t \in [0, \, 1]$

Three coupled nonlinear ODEs — compact, but a genuinely demanding benchmark for any solver.



Bounded, quasi-periodic orbits over $t \in [0, 50]$.

Research Gap & Questions

THE GAP

Prior work lacks a systematic study of plain-ANN capabilities on Lorenz-1960; literature fixes architecture, uses simpler ODEs, or relies on PINNs.

- No architecture study for Lorenz-1960
- No PINN vs plain-ANN comparison
- Depth & width study is rarely combined

OUR QUESTIONS

Can plain neural networks learn coupled chaotic ODE dynamics without physical guidance, and what is the optimal design?

- Can a plain ANN learn the solution from data alone?
- Which configuration wins the accuracy-cost trade-off?
- How close does it get to physics-informed baselines?

Methodology — Two-Phase Study

69 controlled runs

Phase 1 — Architecture search

depth {1, 2, 3, 4} × width {20, 50, 100} × activation {tanh, ReLU, sigmoid, GELU, Swish}

= 60 runs · Adam fixed

Phase 2 — Optimizer comparison

3 best Phase-1 architectures × {Adam, L-BFGS, SGD + momentum}, everything else held constant.

= 9 runs

69 controlled experiments — splitting the phases cleanly separates **architecture** effects from **optimizer** effects.

Generate

RK4 + SciPy



Preprocess

80/20 split



Train

ANN grid



Evaluate

5 metrics



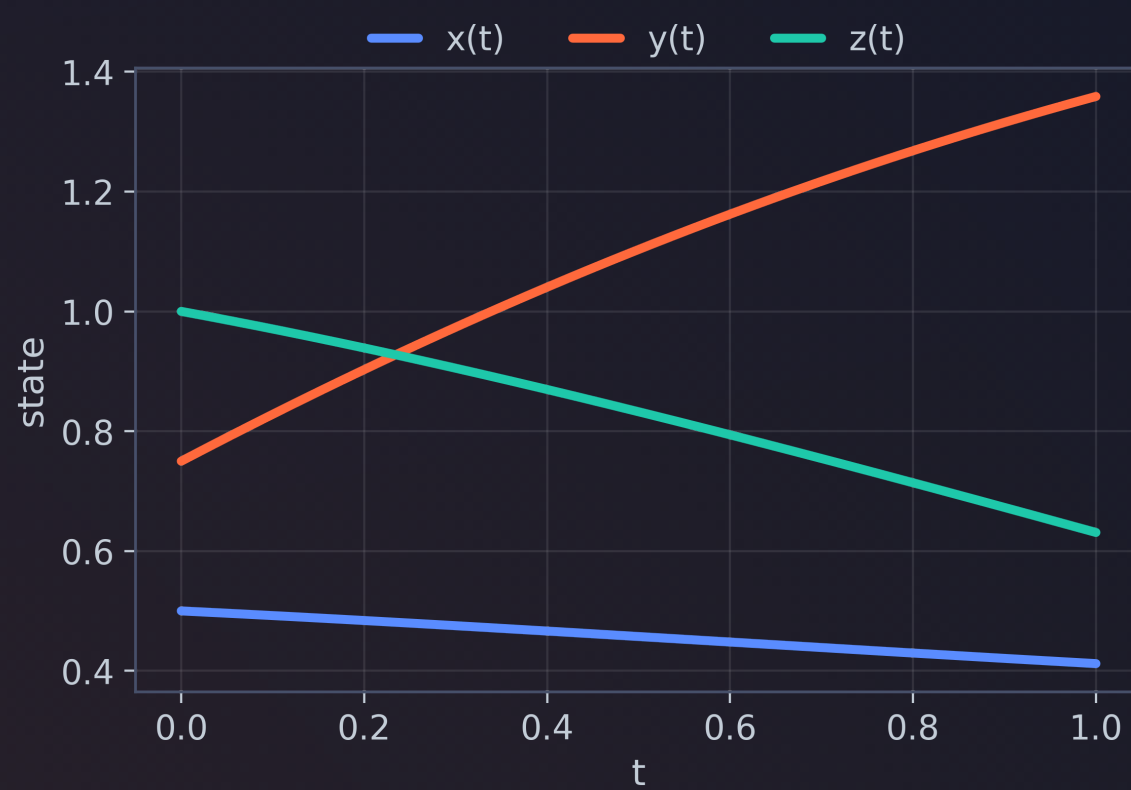
Select

best

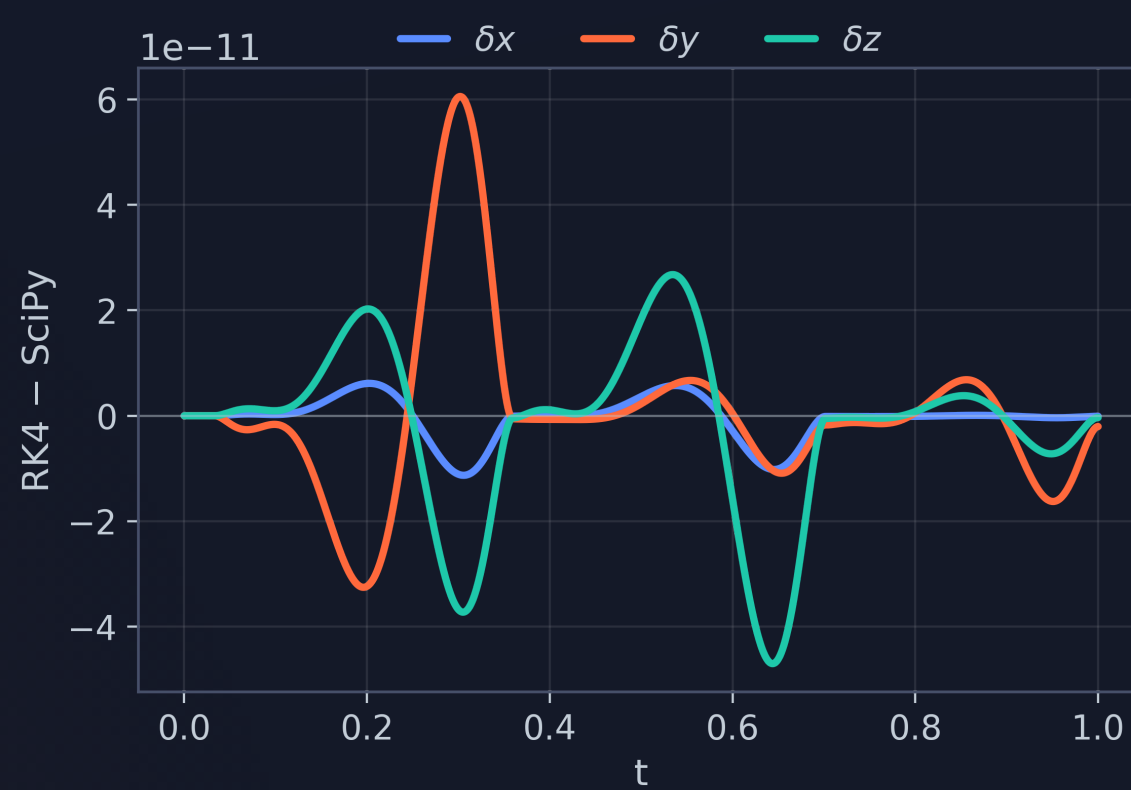
Numerical Baseline & Validation

Result

The reference trajectory is produced by two independent solvers and cross-checked **before any ANN sees it**: a custom **RK4** integrator ($h = 10^{-3}$, 1001 points) and **SciPy DOP853** ($\text{rtol} = 10^{-10}$), split 80/20 for train-validation (seed 42).



Reference Trajectory $t \in [0, \, 1]$



Pointwise Solver Disagreement

 1.8×10^{-22}

SOLVER-AGREEMENT MSE

✓ target was $< 10^{-10}$

A defensible ground truth for ANN training.

Status (FYDP-I) & Future Work

Delivered This Stage (FYDP-I)

- ✓ **Scope Definition:** Research question framed with an honest, narrow focus.
- ✓ **Literature Review:** 17-paper review identifying key architectural gaps.
- ✓ **Experimental Design:** Two-phase design (69 controlled runs) locked.
- ✓ **Numerical Validation:** Dual-solver baseline verified (DOP853 vs RK4).

Next Steps (FYDP-II)

- ➡ **Pipeline Execution:** Run Phase 1 (60 runs) followed by Phase 2 (9 runs).
- ➡ **Stability Testing:** Repeat seeds; dense grids & various initial conditions.
- ➡ **PINN Comparison:** Compare plain ANN directly with physics-informed models.

No accuracy claims are made until the FYDP-II experiments produce evidence — the foundation is set, the results come next.

Selected References

[1] Lorenz (1960), *Tellus* 12(3).

[2] Matthews & Bihlo (2024), PinnDE.

[3] Raissi et al. (2019), *JCP* 378.

[4] Lagaris et al. (1998), *IEEE TNN* 9(5).

[5] Lau & Werth (2020), ODEN.

[6] Chen et al. (2018), Neural ODEs, *NeurIPS*.

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github.com/ihmorol/fydp_workspace